Quantum Protocol & Algorithm Simulator

An interactive, physics-accurate platform for quantum computing education

Version 1.0 • 2026

Agenda

- 1. What is a Quantum Simulator?
- 2. System Architecture
- 3. The Quantum State — Qubits, Superposition & Hilbert Space
- 4. Complete Gate Library (17 gates, all physics-verified)
- 5. Measurement & Collapse
- 6. Bloch Sphere Visualization
- 7. Entanglement & Bell States
- 8. Algorithm: Deutsch-Jozsa
- 9. Algorithm: Grover’s Search
- 10. Protocol: Quantum Teleportation
- 11. Protocol: BB84 Quantum Key Distribution
- 12. Quantum Random Number Generator
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What is the Quantum Simulator?

- Full statevector simulation of 1–5 qubit quantum systems
- 17 physics-verified unitary gates (all satisfy $U^\dagger U = I$)
- Step-by-step circuit execution with time-travel scrubbing
- Real-time 3D Bloch sphere visualization per qubit
- 5 pre-built algorithms & cryptographic protocols
- WebSocket-powered live state updates

Tech Stack

- Backend: Python • FastAPI • NumPy • Socket.IO
- Frontend: React • TypeScript • Three.js • Zustand

Key Principle

- Every gate matrix is explicit — no black-box libraries.
- Physics first: all math is transparent and verifiable.

System Architecture

Frontend (React + TypeScript)

- AlgorithmSelector — configure & run algorithms
- CircuitEditor — drag-and-drop gate placement
- BlochSphere — 3D WebGL per-qubit visualization
- StatevectorPanel — amplitude/probability tables
- Zustand store — centralized state management

Backend (FastAPI + NumPy)

- QuantumState — statevector engine (2^n amplitudes)
- Gates module — 17 unitary matrices
- Measurement — projective collapse & renormalization
- Algorithms — D-J, Grover, Teleport, BB84, QRNG
- Socket.IO — real-time step-by-step push

⇔ **WebSocket / REST API** ⇔

API Endpoints

- GET /api/gates — list all gates | GET /api/algorithms — list algorithms
- POST /api/simulate — run custom circuit | POST /api/algorithms/{name}/run
- Socket.IO: start_session → session_started → step → step_update

The Quantum State

Qubits, Superposition & Hilbert Space

Qubits & Superposition

A qubit is a two-level quantum system. Unlike a classical bit (0 or 1), a qubit exists in superposition:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad \text{where } \alpha, \beta \in \mathbb{C} \text{ and } |\alpha|^2 + |\beta|^2 = 1$$

For n qubits, the state lives in a 2^n-dimensional complex Hilbert space:

$$|\psi\rangle = \sum c_i|i\rangle \quad (2^n \text{ complex amplitudes}, \sum|c_i|^2 = 1)$$

Simulator Implementation

- State stored as numpy complex128 array of 2^n entries
- Big-endian ordering: q_0 = most significant bit
- Max 5 qubits = 32 amplitudes
- Auto-renormalization after every gate

Example: 2 Qubits

- $|00\rangle \rightarrow$ index 0, $|01\rangle \rightarrow$ index 1
- $|10\rangle \rightarrow$ index 2, $|11\rangle \rightarrow$ index 3
- Bell state: [1/√2, 0, 0, 1/√2]
- Probabilities: P(00)=0.5, P(11)=0.5

Single-Qubit Gates (Fixed)

I	`[[1,0],[0,1]]`	Identity — no-op
X	`[[0,1],[1,0]]`	Pauli-X (NOT) — bit flip: $\|0\rangle \leftrightarrow \|1\rangle$
Y	`[[0,-i],[i,0]]`	Pauli-Y — bit+phase flip
Z	`[[1,0],[0,-1]]`	Pauli-Z — phase flip: $\|1\rangle \rightarrow -\|1\rangle$
H	`1/√2 [[1,1],[1,-1]]`	Hadamard — superposition gate
S	`[[1,0],[0,i]]`	$S = \sqrt{Z} = P(\pi/2)$
S†	`[[1,0],[0,-i]]`	S-dagger $= P(-\pi/2)$
T	`[[1,0],[0,e^(iπ/4)]]`	$T = \sqrt{S} = P(\pi/4)$
T†	`[[1,0],[0,e^(-iπ/4)]]`	T-dagger $= P(-\pi/4)$

Key Identities (verified by unit tests)

- $XX = YY = ZZ = HH = I$ | $HXH = Z$, $HZH = X$ | $SS^\dagger = TT^\dagger = I$ | $S^2 = Z$, $T^2 = S$

Parameterized Rotation Gates

P(ϕ)

```
[[1, 0], [0, e^(iϕ)]]
```

General phase gate. Applies e^(iϕ) to |1⟩.
Special cases: P(π/2)=S, P(π/4)=T, P(π)=Z

Ry(θ)

```
[[cosθ/2, -sinθ/2],
 [sinθ/2, cosθ/2]]
```

Rotation around Y-axis by θ radians.
Only real entries — rotates between |0⟩ and |1⟩ without phases.

Rx(θ)

```
[[cosθ/2, -i·sinθ/2],
 [-i·sinθ/2, cosθ/2]]
```

Rotation around X-axis by θ radians.
Traces a great circle through |0⟩ and |1⟩ on the Bloch sphere.

Rz(θ)

```
[[e^(-iθ/2), 0],
 [0, e^(iθ/2)]]
```

Rotation around Z-axis by θ radians.
Bloch vector rotates in the XY equatorial plane.

Multi-Qubit Gates & Entanglement

CNOT (CX)

2 qubits

Controlled-NOT: flips target if control = $|1\rangle$.
The primary entangling gate.
H(q0) + CNOT(q0,q1) → Bell state $(|00\rangle+|11\rangle)/\sqrt{2}$
Matrix:
[[1,0,0,0],[0,1,0,0],[0,0,0,1],[0,0,1,0]]

CZ

2 qubits

Controlled-Z: applies Z to target if control = $|1\rangle$.
Symmetric: CZ(a,b) = CZ(b,a).
Matrix: diag(1, 1, 1, -1)

SWAP

2 qubits

Exchanges two qubit states:
$|01\rangle \leftrightarrow |10\rangle$.
Decomposable into 3 CNOT gates.
Matrix swaps rows/cols for $|01\rangle$ and $|10\rangle$

Toffoli (CCX)

3 qubits

Flips target only when BOTH controls = $|1\rangle$.
Universal for classical computation.
$|110\rangle \leftrightarrow |111\rangle$ (8×8 matrix)

Measurement & Collapse

The Born Rule & Projective Measurement

Quantum Measurement

The Born Rule

- Probability of outcome i: $P(i) = |\langle i|\psi\rangle|^2 = |c_i|^2$
- Post-measurement state: $|\psi'\rangle = P_i|\psi\rangle / \sqrt{P(i)}$
- Measurement is irreversible — information is lost
- The outcome is fundamentally random

Simulator Implementation

- Calculate P(0) by summing $|c_i|^2$ for basis states with bit $j = 0$
- Draw random $r \in [0,1)$: outcome = 0 if $r < P(0)$, else 1
- Project: zero out amplitudes inconsistent with outcome
- Renormalize: divide by $\sqrt{P(\text{outcome})}$

Measurement Bases (used in BB84)

- Z-basis (computational): $|0\rangle$ / $|1\rangle$ — standard measurement, no transformation needed
- X-basis (Hadamard): $|+\rangle$ / $|-\rangle$ — apply H before measuring in Z to rotate $|+\rangle \to |0\rangle$, $|-\rangle \to |1\rangle$
- Y-basis: $|+i\rangle$ / $|-i\rangle$ — apply $S^\dagger$ then H before measuring in Z
- After basis transformation, a standard Z-measurement gives the result in the chosen basis

Bloch Sphere Visualization

Bloch Vector from Density Matrix

- Reduced density matrix: $\rho = Tr_B(|\psi\rangle\langle\psi|)$
- $x = 2\cdot Re(\rho_{01}) = Tr(\rho\cdot\sigma_x)$
- $y = -2\cdot Im(\rho_{01}) = Tr(\rho\cdot\sigma_y)$
- $z = \rho_{00} - \rho_{11} = Tr(\rho\cdot\sigma_z)$
- Purity $r = \sqrt{x^2+y^2+z^2}$: 1.0 = pure, <1.0 = mixed

Key Positions on the Sphere

- $|0\rangle \rightarrow$ North pole (0, 0, +1)
- $|1\rangle \rightarrow$ South pole (0, 0, −1)
- $|+\rangle \rightarrow$ +X equator (+1, 0, 0)
- $|-\rangle \rightarrow$ −X equator (−1, 0, 0)
- $|+i\rangle \rightarrow$ +Y equator (0, +1, 0)
- $|-i\rangle \rightarrow$ −Y equator (0, −1, 0)

What the Bloch Sphere Tells You

- Pure qubit: vector on the surface (r = 1). Gate rotations trace paths on the sphere.
- Entangled qubit: vector inside the sphere (r < 1). Information is stored in correlations, not locally.
- Hadamard: north pole → equator (+X). Pauli-X: north → south (180° around X-axis).
- Rotation gates Rx/Ry/Rz: continuous rotation around the named axis by angle θ.

Entanglement & Bell States

Two qubits are entangled when their joint state cannot be written as a tensor product $|a\rangle \otimes |b\rangle$.
The four Bell states form a maximally entangled basis for two qubits:

$|\Phi^+\rangle$

$(|00\rangle + |11\rangle) / \sqrt{2}$

Circuit:
H(q0), CNOT(q0,q1)

$|\Phi^-\rangle$

$(|00\rangle - |11\rangle) / \sqrt{2}$

Circuit:
H(q0), Z(q0), CNOT(q0,q1)

$|\Psi^+\rangle$

$(|01\rangle + |10\rangle) / \sqrt{2}$

Circuit:
H(q0), CNOT(q0,q1), X(q1)

$|\Psi^-\rangle$

$(|01\rangle - |10\rangle) / \sqrt{2}$

Circuit:
H(q0), Z(q0), CNOT(q0,q1), X(q1)

No-Cloning Theorem

- It is physically impossible to copy an arbitrary unknown quantum state.
- This is why teleportation destroys the original, BB84 is secure, and quantum info can't be broadcast.
- Proof: if $U|0\rangle|\psi\rangle = |\psi\rangle|\psi\rangle$ for all $|\psi\rangle$, then U is not unitary — contradiction.

Deutsch-Jozsa Algorithm

Exponential speedup for oracle classification

Deutsch-Jozsa Algorithm

The Problem

- Given $f: \{0,1\}^n \rightarrow \{0,1\}$, promised to be constant or balanced
- Classical: needs $2^{n-1} + 1$ queries (worst case)
- Quantum: exactly 1 query!
- Exponential speedup via quantum interference

Circuit Steps

- 1. Prepare $|0\rangle^n|1\rangle$ (n input qubits + 1 ancilla)
- 2. Apply H to all qubits (uniform superposition)
- 3. Apply oracle U_f (phase kickback: $|x\rangle \rightarrow (-1)^{f(x)}|x\rangle$)
- 4. Apply H to input qubits
- 5. Measure: all $|0\rangle$ = constant, any $|1\rangle$ = balanced

Physics: Why It Works — Quantum Interference

- The ancilla in $|-\rangle$ enables phase kickback: the oracle encodes f(x) as a ±1 phase on each $|x\rangle$.
- For constant f: all amplitudes get the same phase → final Hadamard recombines them to $|0\rangle^n$ (constructive).
- For balanced f: half get +1, half get −1 → final Hadamard causes destructive interference at $|0\rangle^n$.
- Result: measurement distinguishes constant vs balanced with 100% certainty in a single query.
- Simulator oracle types: constant_0 (identity), constant_1 (X on ancilla), balanced (f(x) = popcount(x & pattern) mod 2).

Grover’s Search Algorithm

Quadratic speedup for unstructured search

Grover's Search Algorithm

The Problem

- Search unsorted database of $N = 2^n$ items
- Classical: $O(N)$ queries
- Quantum: $O(\sqrt{N})$ queries — quadratic speedup

Optimal Iterations

- $k = \lfloor \pi/4 \cdot \sqrt{(N/M)} \rfloor$
- N = search space size (2^n), M = number of targets
- Too many iterations → probability decreases (overshooting)

Amplitude Amplification — The Core Physics

- 1. Initialize: $H^{\otimes n}|0\rangle^n = |s\rangle$ (uniform superposition, all amplitudes = $1/\sqrt{N}$)
- 2. Oracle: flip phase of target states — $|t\rangle \rightarrow -|t\rangle$ (diagonal matrix with −1 at target indices)
- 3. Diffusion: reflect about the mean — operator $2|s\rangle\langle s| - I$ (inversion about average amplitude)
- 4. Geometric picture: state vector lives in 2D subspace ($|\text{target}\rangle$, $|\text{non-target}\rangle$). Each iteration rotates by 2θ where $\sin\theta = \sqrt{(M/N)}$.
- 5. After $\sim\pi/(4\theta)$ iterations, state aligns with target subspace → measure with probability ≈ 1.
- In the simulator: watch probability histogram — target bar grows with each iteration while others shrink.

Quantum Teleportation

Transferring quantum states using entanglement + classical communication

Quantum Teleportation Protocol

1

1. Bell Pair

H(q1), CNOT(q1,q2)
q1 and q2 now share
$|\Phi+\rangle = (|00\rangle+|11\rangle)/\sqrt{2}$

2

2. Alice Acts

CNOT(q0,q1), H(q0)
Entangles message qubit
with her Bell half

3

3. Alice Measures

Measure q0 and q1
Obtains 2 classical bits
(m0, m1)

4

4. Bob Corrects

If m1=1: apply X
If m0=1: apply Z
q2 is now in state $|\psi\rangle$

Physics: Why Teleportation Works

- After Alice's CNOT+H, the 3-qubit state decomposes into 4 terms conditioned on Alice's measurement outcomes.
- Each outcome leaves Bob's qubit in a known Pauli rotation of the original $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$.
- Classical bits (m0, m1) tell Bob which correction to apply: I, X, Z, or XZ.
- No faster-than-light communication: Bob's qubit is random until he receives Alice's classical bits.
- The original state is destroyed (no-cloning): Alice's qubit collapses upon measurement.
- Simulator: set θ/ϕ sliders for q0, run protocol, verify Bob's Bloch sphere (q2) matches the original.

BB84 Quantum Key Distribution

Provably secure cryptography from quantum mechanics

BB84 Protocol

Protocol Steps

- 1. Alice generates random bits, picks random basis (Z or X) each
- 2. Z-basis: $0 \rightarrow |0\rangle$, $1 \rightarrow |1\rangle$. X-basis: $0 \rightarrow |+\rangle$, $1 \rightarrow |-\rangle$
- 3. Bob picks random measurement basis per qubit
- 4. Sifting: keep bits where bases matched (~50%)
- 5. Estimate QBER on a sacrificed subset
- 6. QBER < 11% → secure. QBER ≥ 11% → Eve detected!

Why Eve Gets Caught

- Eve must measure to read — but doesn't know Alice's basis
- Wrong basis guess (50%): measurement collapses to wrong basis
- Re-sent qubit: Bob gets error 50% of the time (when matching Alice)
- Net effect: Eve introduces ~25% QBER
- Security threshold: 11% (provably secure bound for BB84)

Security Foundation — Quantum Physics Guarantees

- No-Cloning Theorem: Eve cannot copy qubits to measure in both bases simultaneously.
- Measurement Disturbance: any measurement on a qubit in a non-matching basis irreversibly disturbs it.
- Information-theoretic security: security does not depend on computational assumptions — it comes from physics.
- Simulator: toggle Eve on/off and observe QBER jump from ~0% to ~25%.

Quantum Random Number Generator (QRNG)

The Concept

- Classical PRNGs are deterministic (predictable with seed)
- Quantum mechanics provides TRUE randomness
- Measuring a superposition is fundamentally unpredictable
- No hidden variables (Bell's theorem)

The Circuit (per bit)

- 1. Prepare fresh qubit: $|0\rangle$
- 2. Apply Hadamard: $|0\rangle \rightarrow (|0\rangle+|1\rangle)/\sqrt{2}$
- 3. Measure: exactly 50/50 probability
- 4. Repeat N times for N truly random bits

Simulator Output

- Bitstring: the generated random binary sequence (up to 256 bits)
- Integer / Hex value: decimal and hexadecimal representation (≤64 bits)
- Frequency analysis: count of 0s vs 1s — should approach 50/50 for large N
- State history: step-by-step statevector at each H → Measure cycle

Live Demo Walkthrough

Demo 1: Superposition

- 1 qubit system
- Apply H to q0
- Observe Bloch sphere: north pole → equator
- Probability chart: 50/50 split

Demo 2: Bell State

- 2 qubit system
- Apply H(q0), then CNOT(q0,q1)
- Both Bloch spheres shrink (entangled)
- Probabilities: only $|00\rangle$ and $|11\rangle$

Demo 3: Grover's Search

- 3 qubits, target = 5 ($|101\rangle$)
- Watch amplitude amplification
- Target probability rises each iteration
- Final measurement finds target

Demo 4: Teleportation

- Set arbitrary state on q0
- Run protocol, step through
- Observe Bob's q2 matches original
- Original q0 is destroyed (no-cloning)

Tips for Exploration

- Use Step Forward/Backward to freeze time and inspect states between gates.
- Compare Deutsch-Jozsa with constant_0 vs balanced — see the interference patterns differ.
- Toggle Eve in BB84: watch QBER jump from ~0% to ~25%.
- Try QRNG with 200 bits and verify frequency approaches 50/50.

Thank You

Quantum Protocol & Algorithm Simulator

Questions?